

Rajnish Kumar

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Professional Summary

GenAI Engineer with 2+ years of experience building enterprise Generative AI solutions using Python, LLMs, RAG, LangChain, and LangGraph. Experienced in developing AI assistants and agentic workflows with retrieval, tool integration, vector databases, and LLM orchestration for enterprise banking use cases.

Technical Skills

Programming: Python, SQL

Generative AI: LLMs, RAG, AI Agents, Agentic AI, Multi-Agent Systems, Prompt Engineering, Tool Calling, Function Calling, LLM Evaluation

LLM & Frameworks: Gemini, Groq, Cerebras, LangChain, LangGraph, HuggingFace Transformers, Sentence Transformers

RAG & Vector Search: Embeddings, Semantic Search, Document Ingestion, Text Chunking, Similarity Search, Top-K Retrieval, ChromaDB

AI Engineering: LLM Orchestration, Agent Workflows, Memory, State Management, API Integration, Response Validation, Caching

Backend & DevOps: FastAPI, REST APIs, Pydantic, Docker, AWS EC2, Jenkins, GitHub Actions, CI/CD

Tools: Git, GitHub, Linux

Professional Experience

LTIMindtree

Jun 2024 – Present

Software Engineer — Generative AI

- Developed enterprise GenAI solutions for banking use cases using Python, LLMs, RAG, LangChain, and LangGraph.
- Built a **GenAI Operations & Incident Assistant** using LLMs and RAG to retrieve relevant operational knowledge, analyze incident context, summarize issues, and generate troubleshooting recommendations.
- Developed an **AI Event & Incident Triage Agent** using agentic workflows to analyze incoming events, retrieve historical incidents and technical documentation, and generate incident analysis and recommended actions.
- Designed RAG pipelines covering document ingestion, text chunking, embeddings, semantic retrieval, vector search, context construction, and LLM-based response generation.
- Implemented LangGraph-based workflows with state management, conditional routing, tool integration, and multi-step LLM orchestration.
- Developed Python-based backend services and REST APIs using FastAPI, implementing structured outputs, response validation, error handling, and performance optimization for GenAI applications.

LTIMindtree

Feb 2023 – Apr 2023

Cloud Engineer Intern

- Assisted in provisioning AWS cloud infrastructure and deploying containerized applications using Docker.
- Supported deployment automation, infrastructure monitoring, cloud operations, and infrastructure management activities.

Projects

Production-Ready Multi-Agent AI Trading Research Agent

GitHub Repo

Python, FastAPI, LangGraph, LangChain, ChromaDB, Docker, GitHub Actions, AWS EC2

- Designed and developed a production-ready Multi-Agent AI Trading Research Agent using LangGraph, RAG, and LLMs to automate financial market research and report generation.
- Architected modular Retriever, News, Sentiment, Risk, Summary, and Memory agents using ChromaDB and HuggingFace Embeddings for context-aware analysis.
- Integrated Tavily Search, yFinance, and multiple LLM providers (Groq, Gemini, Cerebras) for real-time market data, news analysis, and semantic retrieval.
- Developed FastAPI REST APIs with health checks, Docker containerization, and automated CI/CD deployment to AWS EC2 using GitHub Actions and Docker Hub.
- Engineered reliable deployments using semantic versioning, idempotent deployment scripts, retry-based health checks, and Docker restart policies.
- Optimized performance through embedding reuse, persistent vector storage, and efficient semantic retrieval.

HireSense AI – LLM-Based Resume Intelligence & Ranking Engine

GitHub Repo

Python, OpenAI API, LangChain, FastAPI, Streamlit, Embeddings

- Built an LLM-powered platform for semantic resume-job matching, candidate evaluation, and intelligent resume ranking.
- Developed embedding-based semantic similarity pipelines for resume and job-description matching.
- Implemented AI-driven skill extraction, resume evaluation, and contextual feedback generation using OpenAI models.
- Developed FastAPI REST APIs and optimized AI workflows using embedding reuse, response caching, and retrieval optimization.

Image2Insight – OCR-Based Sentiment Analysis

GitHub Repo

Python, OCR, NLP, VADER Sentiment Analysis

- Developed an OCR-based NLP application to extract textual information from image documents and generate automated insights.
- Implemented image preprocessing and text-processing pipelines to improve OCR output and downstream analysis.
- Applied VADER sentiment analysis to classify extracted text and automate sentiment-based insights.

Leadership & Communication

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- Delivered technical presentations and communicated complex technical concepts to diverse audiences.
 - Collaborated with cross-functional teams to drive knowledge sharing and achieve common objectives.
 - Coordinated team initiatives and supported peer mentoring while strengthening stakeholder communication, active listening, problem-solving, and feedback skills.

Education

Global Academy of Technology

Bengaluru, India

Bachelor of Engineering (B.E.)

Electronics and Communication Engineering

2019 – 2023